

1 Solution — GIAM 2.2

Problem 10

1.1 Problem

The **integral test**: suppose $f(x)$ is a positive, decreasing, real-valued function with $\lim_{x \rightarrow \infty} f(x) = 0$. If the improper integral $\int_0^\infty f(x) dx$ has a finite value, then the infinite series $\sum_{n=1}^\infty f(n)$ converges. The test is envisioned via a right-hand Riemann sum, which (as f decreases) underestimates the integral:

$$\sum_{n=1}^{\infty} f(n) < \int_0^{\infty} f(x) dx.$$

Discuss the meanings of and (where possible) provide justifications for the **inverse**, **converse**, and **contrapositive** of the conditional in the integral test.

1.2 Setting Up

Write the conditional as $P \Rightarrow Q$ with

$$P : \int_0^\infty f(x) dx \text{ converges,} \quad Q : \sum_{n=1}^\infty f(n) \text{ converges,}$$

throughout assuming the standing hypotheses (f positive, decreasing, $f \rightarrow 0$). The four relatives are:

name	form	meaning
original	$P \Rightarrow Q$	integral converges $\Rightarrow$ series converges
contrapositive	$\neg Q \Rightarrow \neg P$	series diverges $\Rightarrow$ integral diverges
converse	$Q \Rightarrow P$	series converges $\Rightarrow$ integral converges
inverse	$\neg P \Rightarrow \neg Q$	integral diverges $\Rightarrow$ series diverges

Table 1.1

All four are true here — but for different reasons, and that is the point of the exercise.

1.3 The Contrapositive — True, and Free

$$\boxed{\text{If } \sum_{n=1}^{\infty} f(n) \text{ diverges, then } \int_0^{\infty} f(x) dx \text{ diverges.}}$$

This needs **no new work**: by Problem 8, a conditional and its contrapositive are logically equivalent, so it is true the moment the original is. It is also the *practically useful* form — it is how one proves an integral diverges by recognising a divergent series (or, read the other way, how a divergent series certifies a divergent integral).

The underlying justification is the same inequality. Since $\sum f(n) \leq \int_0^{\infty} f$, an infinite left side forces an infinite right side.

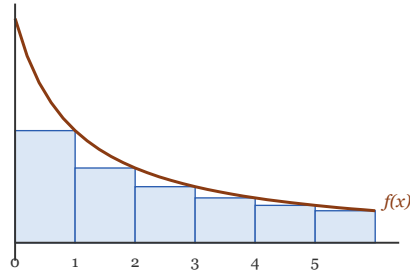
1.4 The Converse — Also True, but Needs a *New* Argument

$$\boxed{\text{If } \sum_{n=1}^{\infty} f(n) \text{ converges, then } \int_0^{\infty} f(x) dx \text{ converges.}}$$

This does **not** follow from the original (converses never do — Problem 9). It happens to be true, but requires the *other* Riemann sum: the **left-hand** sum, which for a decreasing f overestimates.

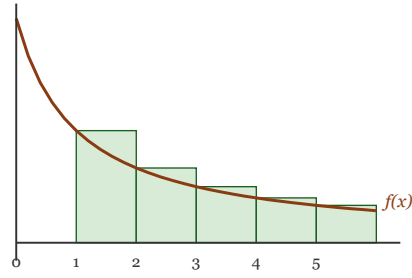
Two Riemann sums bracket the integral. The right-hand sum proves the original; the left-hand sum proves the converse.

Right-hand rectangles — UNDER the curve
 $\sum f(n) \leq \int_0^\infty f(x) dx$



each bar of height $f(n)$ fits inside its strip,
 so the total bar area is at most the area under f .
integral finite $\Rightarrow$ series finite

Left-hand rectangles — OVER the curve
 $\int_1^\infty f(x) dx \leq \sum f(n)$



each bar of height $f(n)$ covers its strip,
 so the total bar area is at least the area under f on $[1, \infty)$.
series finite $\Rightarrow$ integral finite (the converse)

Figure 1.1: Two side-by-side panels, each showing a positive decreasing curve $f(x)$ with unit-width rectangles. Left panel, right-hand rectangles: on each strip from $n-1$ to n the bar has height $f(n)$ and fits entirely under the curve, so the total bar area is at most the area under f — giving sum of $f(n)$ at most the integral from 0 to infinity. This proves ‘integral finite implies series finite’, the original. Right panel, left-hand rectangles: on each strip from n to $n+1$ the bar has height $f(n)$ and covers the curve, so the total bar area is at least the area under f on $[1, \infty)$ — giving the integral from 1 to infinity at most the sum. This proves ‘series finite implies integral finite’, the converse.

Justification. On the strip $[n, n+1]$, decreasing f means $f(x) \leq f(n)$ for all x there, so $\int_n^{n+1} f(x) dx \leq f(n)$. Summing over $n \geq 1$:

$$\int_1^\infty f(x) dx \leq \sum_{n=1}^\infty f(n).$$

Now suppose the series converges, say to $S < \infty$. Then $\int_1^\infty f \leq S$ is finite. The leftover piece is finite too: f is decreasing and positive, so on $[0, 1]$ it is bounded by $f(0)$, giving $\int_0^1 f(x) dx \leq f(0) < \infty$. Adding,

$$\int_0^\infty f = \int_0^1 f + \int_1^\infty f \leq f(0) + S < \infty. \quad \blacksquare$$

1.5 The Inverse — True, Because It Is the Contrapositive of the Converse

If $\int_0^\infty f(x) dx$ diverges, then $\sum_{n=1}^\infty f(n)$ diverges.
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The inverse $\neg P \Rightarrow \neg Q$ is the contrapositive of the converse $Q \Rightarrow P$, so it is *automatically* true once the converse is established (Problem 9’s observation that converse $\equiv$ inverse). Directly: if $\int_0^\infty f = \infty$ then $\int_1^\infty f = \infty$, and since $\int_1^\infty f \leq \sum f(n)$, the series must diverge too.

1.6 Remark — The Integral Test Is Really a Biconditional

Because the original *and* its converse both hold, the two may be combined (Problem 9):

$$\int_0^\infty f(x) dx \text{ converges} \iff \sum_{n=1}^\infty f(n) \text{ converges.}$$

Under the standing hypotheses, integral and series **live or die together**. This is why the integral test is a genuine *test* rather than a one-way sufficient condition: it can certify convergence *and* divergence. Most textbooks state it as an “if and only if” for exactly this reason. The whole content of the two-sided result is captured by the sandwich

$$\int_1^\infty f(x) dx \leq \sum_{n=1}^\infty f(n) \leq \int_0^\infty f(x) dx,$$

with the left inequality (left-hand sum) giving one direction and the right inequality (right-hand sum) the other.

1.7 Remark — Why the Converse Genuinely Needed New Work

It is worth stressing that we could *not* have obtained the converse for free. Logic guarantees only the contrapositive; the converse is an independent claim (Problem 9), and here it demanded a second geometric argument with the rectangles shifted one unit. That the answer came out “true” is a fact about *this particular* f -hypothesis, not about conditionals in general. Had we only ever drawn the right-hand rectangles, we would possess a one-directional test — able to prove convergence but never divergence — which is exactly half as useful.

1.8 Remark — Where the Hypotheses Earn Their Keep

Each standing assumption does real work, and dropping any one breaks the equivalence:

- **Decreasing** is essential to both inequalities: it is what puts $f(n)$ below f on $[n-1, n]$ and above f on $[n, n+1]$. For a non-monotone f (say one with tall thin spikes at each integer), the series can diverge while the integral converges — the spikes have negligible area but huge sampled values.
- **Positive** ensures the partial sums and partial integrals are increasing, so “converges” means “bounded”; with sign changes, cancellation could make an integral converge conditionally while the sampled series does not.
- $f \rightarrow 0$ is implied by the others in the convergent case, but stating it flags the necessary condition for any series to converge.

1.9 Remark — A Worked Instance: The p -Series

The biconditional is what makes the p -series verdict so clean. Take $f(x) = 1/(x+1)^p$, positive and decreasing for $p > 0$. The integral $\int_0^\infty (x+1)^{-p} dx$ converges exactly when $p > 1$, so by the two-way test $\sum 1/n^p$ converges exactly when $p > 1$. Both halves are used: for $p > 1$ the *original* (finite integral $\Rightarrow$ finite sum) settles convergence; for $p \leq 1$ the *inverse* (infinite integral $\Rightarrow$ infinite sum) settles divergence — including the harmonic series $p = 1$, whose divergence is certified by $\int_1^\infty dx/x = \infty$. A test that only ran one direction could prove half of this and no more.